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Cloud Computing Assignment

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This project helped me understand how machine learning systems can be integrated with cloud computing infrastructure to build and deploy a complete application. Before this project, I had worked with machine learning models and backend development, but I had not implemented an end to end pipeline involving teacher generated data, model distillation, object storage, containerisation, Kubernetes deployment, autoscaling and performance evaluation.

One of my biggest learnings was the importance of data quality in knowledge distillation. I used Phi-3.5 as the teacher model to generate labels for 2000 commands using two parallel workers. The labelling process took approx. seven and half hours. Initially the strict parsing pipeline retained only 827 records because many teacher responses contained valid JSON followed by additional text. I investigated these outputs and implemented a seperate recovery process using JSON decoding to extract the ~~first~~ first valid JSON object before applying the original validation rules. This recovered 2821 records. I learned that data-processing decisions can significantly affect the amount of usable training data.

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Another important learning was sequence-level knowledge distillation. I fine tuned Pythia-70M using teacher generated responses and compared two training configurations. The student models achieved 99% and 100% JSON validity, respectively compared with 97% for the teacher on the validation set. However the full frame accuracies were only 46% and 44%. This showed me that generating valid structured o/p is different from correctly extracting every field.

The infrastructure portion of the project was equally valuable. I deployed seaweedFS and s3 compatible object storage and used Kubernetes namespaces to separate data, pipeline and application components.

I faced several challenges during deployment & testing. CPU-Based inference was relatively slow, and concurrent requests pushed Pod CPU utilisation close to one full core. I also encountered malformed model output for a validation example containing the location 'quadcities'.

I also tested rolling updates, rollback & Pod recovery. The application successfully switched from model v_1 to v_2 and back to v_1. Kubernetes recreated a deliberately deleted Pod while the other replica remained healthy.

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If I were to improve this project further, I would collect a larger and more diverse training dataset, introduce stronger semantic validation, investigate constrained decoding, and explore more efficient CPU inference techniques such as quantization. I would also improve traffic draining during rolling updates and test the service on a multi-node cluster.